Homework #4

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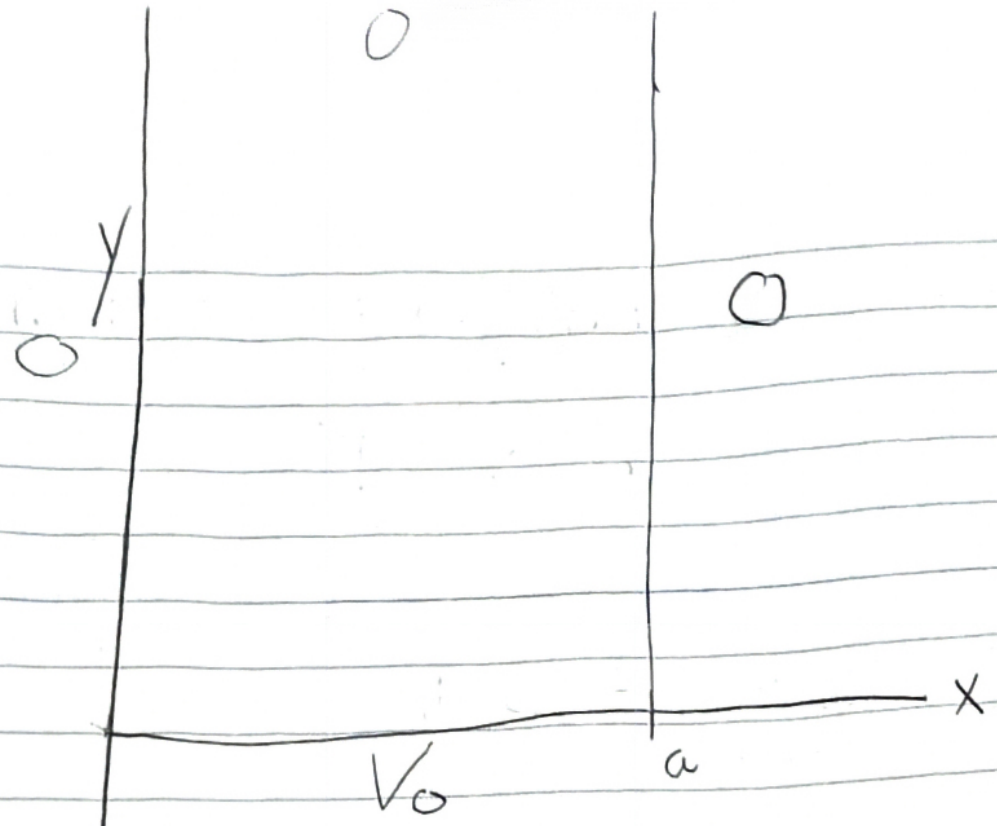
Probl.)

- $0 \leq x \leq a$.
- $y \geq 0$.

a.) Boundary Conditions:

* $\Phi(x,0) = V_0$ (Constant)
* $\Phi(0,y) = 0$
* $\Phi(a,y) = 0$
* $\Phi(x, y \to \infty) = 0$

- $\nabla^2 \phi = 0$
- $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$



- $\Phi(x,y) = A(x)B(y)$

X-equation:

- $\frac{1}{A}\frac{\partial^2 A}{\partial x^2} + \frac{1}{B}\frac{\partial^2 B}{\partial y^2} = 0$

- $\frac{1}{A}\frac{\partial^2 A}{\partial x^2} = -\lambda^2$

$A = e^{rx}$

$A' = re^{rx}$

$A'' = r^2 e^{rx}$

- $A'' + A\lambda^2 = 0$

$r^2 + \lambda^2 = 0, \quad r = \pm\lambda i$

- $A(x) = P\cos(\lambda x) + C\sin(\lambda x)$
  - apply $\phi(0) = 0$
    $\phi(a) = 0$
- $0 = P(1) + C(0)$
- $P = 0$
- $0 = \sin(\lambda a)$
- $\lambda_n a = n\pi$, $\lambda_n = \frac{n\pi}{a}$
- $A_n(x) = C_n \sin\left(\frac{n\pi x}{a}\right)$

- $B = e^{rY}$

Y-equation:

- $\frac{1}{B}\frac{\partial^2 B}{\partial y^2} = \lambda^2$, $B'' - B\lambda^2 = 0$
- Propose Real exponentials where $r = \lambda, -\lambda$

$$B_n(y) = D_n e^{\frac{n\pi y}{a}} + E_n e^{-\frac{n\pi y}{a}}$$

apply Boundary Condition:

- $\Phi(x, y \to \infty) = 0$
- $D_n e^{n\pi y/a}$ Diverges when applying this so drop this term.
- $B_n(y) = L_n e^{\frac{-n\pi y}{a}}$

$* \left(A_n = L_n C_n\right)$

- $\Phi(x,y) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{a}\right) e^{\frac{-n\pi y}{a}}$
- $\Phi(x,0) = V_0$
- $V_0 = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{a}\right)$
- multiply $\int_0^a \sin\left(\frac{m\pi x}{a}\right) dx$

$$V_0 \int_0^a \sin\left(\frac{m\pi x}{a}\right) dx = \sum_{n=1}^{\infty} A_n \int_0^a \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi x}{a}\right) dx$$

- $A_n \frac{a}{2} \delta_{mn} = V_0 \int_0^a \sin\left(\frac{n\pi x}{a}\right) dx$ $\quad \frac{a}{2}\delta_{mn}$

$(m = n)$

- $A_n = \frac{2V_0}{a}\int_0^a \sin\left(\frac{n\pi x}{a}\right) dx$

$u = \frac{n\pi x}{a}$

$du = \frac{n\pi}{a} dx$

- $A_n = \frac{2V_0}{n\pi}\int_0^{n\pi} \sin(u)\, du$

- $A_n = \frac{2V_0}{n\pi}\left[-\cos(u)\right]_0^{n\pi}$

- $A_n = \frac{2V_0}{n\pi}\left[-\cos(n\pi) + \cos(0)\right]$

- $A_n = \begin{cases} \frac{4V_0}{n\pi} & n = \text{odd} \\ 0 & n = \text{even} \end{cases}$

$$\Phi(x,y) = \sum_{n=1,3,5,7,\ldots}^{\infty} \frac{4V_0}{n\pi} \sin\left(\frac{n\pi x}{a}\right) e^{\frac{-n\pi y}{a}}$$

b.)

$$\Phi(x,y) = \frac{4V_0}{\pi} \sum_{n\,\text{odd}} \frac{1}{n} \exp\left(\frac{-n\pi y}{a}\right) \sin\left(\frac{n\pi x}{a}\right)$$

$$\sin\theta = \text{Im}\left(e^{i\theta}\right)$$

$$\Phi(x,y) = \frac{4V_0}{\pi} \text{Im}\left(\sum_{n\,\text{odd}} \frac{1}{n} e^{(in\pi/a)(x+iy)}\right)$$

$$z = e^{(i\pi/a)(x+iy)} = e^{i\pi x/a} e^{-\pi y/a}$$

$$\Phi(x,y) = \frac{4V_0}{\pi} \text{Im}\left(\sum_{n,\text{odd}} \frac{z^n}{n}\right)$$

$$\frac{d}{dz}\left(\sum_{n=1}^{\infty} \frac{z^n}{n}\right) = \sum_{n=0}^{\infty} z^n = \frac{1}{(1-z)}$$

Then integrate both sides

$$\sum_{n=1} z^n/n = -\ln(1-z)$$

$$\sum_{n\,\text{odd}} \frac{z^n}{n\pi} = \frac{1}{2}\ln\left(\frac{1+z}{1-z}\right)$$

$$\Phi(x,y) = \frac{2V_0}{\pi}\,\mathrm{Im}\left\{\sum_{n\,\text{odd}} \frac{1}{n\pi}\ln\left(\frac{1+z}{1-z}\right)\right\}$$

$$\boxed{\Phi(x,y) = \frac{2V_0}{4\pi}\ln\left(\frac{1+\sin(\pi x/a)\,e^{-\pi y/a}}{1-\sin(\pi x/a)\,e^{-\pi y/a}}\right)}$$

$$\mathrm{Im}\left(\sum_{n=0}^{\infty} a_n z^{n+1}\right) = \mathrm{Im}\left(\frac{a}{(1-z)^2}\right)$$

$$\Phi(x,y) = Im\left(\frac{4V_0}{n^2\pi}\left(\frac{1}{\left(1-e^{\pi/a(ix-y)}\right)^2}\right)\right)$$

$$\Phi(x,y) = \frac{4V_0}{n^2\pi}\left(\frac{1}{\left(1-e^{\frac{-\pi y}{a}}\sin\left(\frac{\pi x}{a}\right)\right)^2}\right)$$

c.)

- $\Phi(x,y) = \sum_{n=odd}^{\infty} \frac{4V_0}{n\pi} \sin\left(\frac{n\pi x}{a}\right) e^{\frac{-n\pi y}{a}}$

- express in <u>euler-form</u>:

- $e^{i\theta} = \cos(\theta) + i\sin(\theta)$

- $\Phi(x,y) = \sum_{n=odd}^{\infty} \frac{4V_0}{n\pi} \left(\cos\left(\frac{n\pi x}{a}\right) + i\sin\left(\frac{n\pi x}{a}\right)\right) e^{\frac{-n\pi y}{a}}$

- $\Phi(x,y) = \sum_{n=odd}^{\infty} \frac{4V_0}{n\pi} \cos\left(\frac{n\pi x}{a}\right) e^{\frac{-n\pi y}{a}}$

$$+ \; i \sum_{n=odd}^{\infty} \frac{4V_0}{n\pi} \sin\left(\frac{n\pi x}{a}\right) e^{\frac{-n\pi y}{a}}$$

<u>Cauchy-Riemann conditions:</u>

- $f(z) = u(x,y) + iv(x,y)$
- $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$; $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$

$$u(x,y) = \sum_{n=\text{odd}}^{\infty} \frac{4V_0}{n\pi} \cos\left(\frac{n\pi x}{a}\right) e^{\frac{-n\pi y}{a}}$$

$$v(x,y) = \sum_{n=\text{odd}}^{\infty} \frac{4V_0}{n\pi} \sin\left(\frac{n\pi x}{a}\right) e^{\frac{-n\pi y}{a}}$$

$$\frac{\partial u}{\partial x} = \sum_{n=\text{odd}}^{\infty} -\frac{4V_0}{n\pi}\left(\frac{n\pi}{a}\right) \sin\left(\frac{n\pi x}{a}\right) e^{\frac{-n\pi y}{a}}$$

$$\frac{\partial u}{\partial x} = \sum_{n=\text{odd}}^{\infty} -\frac{4V_0}{a} \sin\left(\frac{n\pi x}{a}\right) e^{\frac{-n\pi y}{a}}$$

$$\frac{\partial v}{\partial y} = \sum_{n=\text{odd}}^{\infty} \frac{-4V_0}{a} \sin\left(\frac{n\pi x}{a}\right) e^{\frac{-n\pi y}{a}}$$

$$\boxed{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}}$$

$$\frac{\partial v}{\partial x} = \sum_{n=\text{odd}}^{\infty} \frac{4V_0}{n\pi}\left(\frac{n\pi}{a}\right) \cos\left(\frac{n\pi x}{a}\right) e^{\frac{-n\pi y}{a}}$$

$$\frac{\partial v}{\partial x} = \sum_{n=\text{odd}}^{\infty} \frac{4V_0}{a} \cos\left(\frac{n\pi x}{a}\right) e^{\frac{-n\pi y}{a}}$$

- $\frac{\partial v}{\partial y} = \sum_{n=odd}^{\infty} \frac{4V_0}{n\pi}\left(\frac{-n\pi}{a}\right)\cos\left(\frac{n\pi x}{a}\right)e^{\frac{-n\pi y}{a}}$

- $\frac{\partial u}{\partial y} = \sum_{n=odd}^{\infty} \frac{-4V_0}{a}\cos\left(\frac{n\pi x}{a}\right)e^{\frac{-n\pi y}{a}}$

- $\frac{-\partial u}{\partial y} = \sum_{n=odd}^{\infty} \frac{4V_0}{a}\cos\left(\frac{n\pi x}{a}\right)e^{\frac{-n\pi y}{a}}$

$$\boxed{\frac{\partial v}{-\partial x} = \frac{-\partial u}{\partial y}}$$

- $\underline{\Phi}(x,y) = \sum_{n=odd}^{\infty} \frac{4V_0}{n\pi} e^{\frac{in\pi x}{a}} e^{\frac{-n\pi y}{a}}$

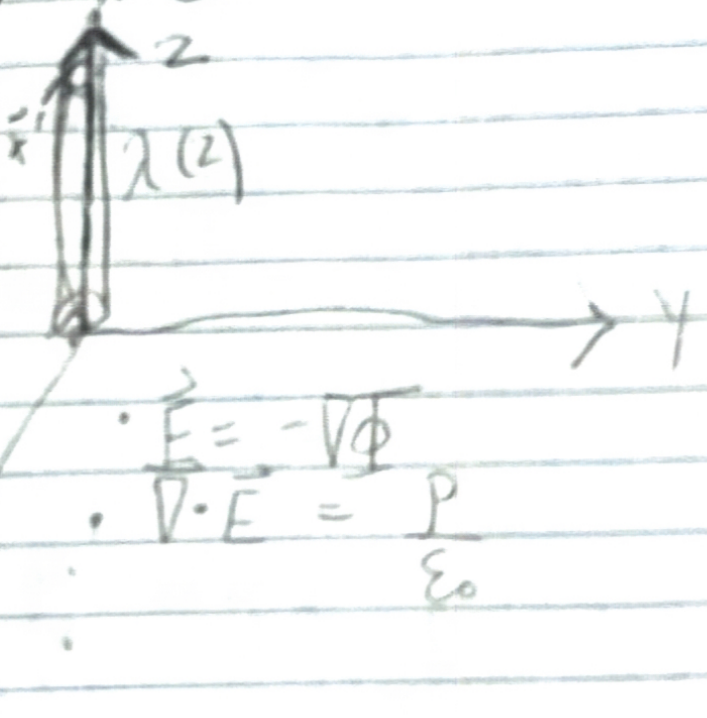
is analytic so therefore

$\underline{\Phi}(x,y) = Im\left(\underline{\Phi}_n(x,y)\right)$ is analytic

2)

- $\lambda(z) = \begin{cases} \lambda_0 \exp(-z/a), & z \geq 0 \\ 0, & z < 0 \end{cases}$

- $\Phi(r \to \infty) = 0$



- $z = r\cos\theta$

- $\vec{E} = -\nabla\Phi$
- $\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$

$$\lambda(r\cos\theta) = \begin{cases} \lambda_0 \exp\left(\frac{-r\cos\theta}{a}\right) & z \geq 0 \\ 0 \end{cases}$$

$$\Phi(\vec{x}) = \frac{1}{4\pi\varepsilon_0}\int_V \frac{\rho(\vec{x}')}{|\vec{x}-\vec{x}'|}\,d^3x'$$

$$0 \le r' \le \infty \qquad 0 \le \theta' \le \pi \qquad 0 \le \phi' \le 2\pi$$

$$\boxed{\rho(\vec{x}') = \lambda_0 \exp\left(\frac{-r'\cos\theta'}{a}\right)\frac{1}{r'}\delta\left(r'-r\right)\delta\left(\theta'-\frac{\pi}{2}\right)}$$

$$\frac{1}{|\vec{x}-\vec{x}'|} = \frac{1}{r_>}\sum_{l=0}^{\infty}\left(\frac{r_<}{r_>}\right)^l$$

($\vec{x}$ is on z-axis)

regions: r

- i) $r > r'$
- ii) $r' < r$

$$\Phi(\vec{x}) = \frac{\lambda_0}{4\pi\varepsilon_0}\int_0^{2\pi}\int_0^{\pi}\int_0^{\infty}\frac{1}{r'}\sum_{l=0}^{\infty}\left(\frac{r_<}{r_>}\right)^l \delta(r-r')\delta(\theta')r'^2\sin\theta'\,dr'd\theta'd\phi'$$

$$= \sum_{l=0}^{\infty}\frac{\lambda_0}{4\pi\varepsilon_0}\int_0^{2\pi}\int_0^{\pi}\int_0^{\infty} r'\left(\frac{r_<}{r_>}\right)^l \delta(r-r')\delta(\theta'-\tfrac{\pi}{2})\sin\theta'\,dr'd\theta'd\phi' \qquad (r'<r')$$

$$= \sum_{l=0}^{\infty}\frac{\lambda_0}{4\pi\varepsilon_0}\int_0^{2\pi}\int_0^{\pi}\int_0^{r+\varepsilon}\frac{r'^l}{r^{l+1}}\delta(r'-r)\,\delta(\theta'-\tfrac{\pi}{2})\sin\theta'\,dr'd\theta'd\phi' \exp\left(\frac{-r'\cos\theta'}{a}\right)$$

$(r' > r)$

$$\sum_{l=0}^{\infty} \frac{\lambda_0}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^{\pi} \int_{r+\epsilon}^{\infty} \frac{r'^{l-1}}{r'^{l}} \delta(r'-r') \sin\theta' dr' d\theta' d\phi' \; \delta(\theta' - \pi/2) \exp(-r'\cos\theta'/a)$$

$$\Phi(\vec{x}) = \sum_{l=0}^{\infty} \frac{\lambda_0}{2\epsilon_0} \left[\int_0^{r+\epsilon} \frac{r'^{l}}{r'^{l+1}} \exp\left(\frac{-r'}{a}\right) dr' \delta(r'-r) + \int_{r+\epsilon}^{\infty} \frac{r^{l-1}}{r'^{l}} \exp\left(\frac{-r'}{a}\right) \delta(r'-r) dr' \right]$$

Prob 3.)

$$\left|\vec{x}-\frac{a}{2}\hat{z}\right|^2 = \left(\vec{x}-\frac{a}{2}\hat{z}\right)\cdot\left(\vec{x}-\frac{a}{2}\hat{z}\right)$$

$$= \vec{x}\cdot\vec{x} - \frac{a}{2}\vec{x}\cdot\hat{z} - \frac{a}{2}\vec{x}\cdot\hat{z} + \frac{a^2}{4}\hat{z}\cdot\hat{z}$$

$$= \vec{x}\cdot\vec{x} + \frac{a^2}{4} - a\left(\vec{x}\cdot\hat{z}\right)$$

$$= \frac{4+a^2}{4} - a\left(\vec{x}\cdot\hat{z}\right)$$

$$\vec{x} = r\sin\theta\cos\phi\,\hat{x} + r\sin\theta\sin\phi\,\hat{y} + r\cos\theta\,\hat{z}$$

if I define $\vec{x} = r\hat{r}$ then $\vec{x}\cdot\vec{x} = r^2$

$$\left|\vec{x}-\frac{a}{2}\hat{z}\right| = \sqrt{r^2+\frac{a^2}{4} - ra\cos\theta}$$

a.) Boundary condition:

$$\Phi(a,\cos\theta) = V_0(1+\cos\theta)$$

$$\Phi_2 = \frac{q}{4\pi\varepsilon_0\left|\vec{x}-a/2\,\hat{z}\right|} = \frac{q}{4\cdot\pi\varepsilon_0\left(r^2+\frac{a^2}{4}-ra\cos\theta\right)^{1/2}}$$

$$\Phi = \frac{q}{4\pi\varepsilon_0\left(r^2+\frac{a^2}{4}-ra\cos\theta\right)^{1/2}} + \Phi_0$$

apply boundary condition

$$\Phi(a, \cos\theta) = V_0(1+\cos\theta)$$

$$V_0(1+\cos\theta) = \frac{q}{4\pi\varepsilon_0\left(\frac{5a^2}{4}-a^2\cos\theta\right)^{1/2}} + \Phi_0$$

$$V_0 + V_0\cos\theta = \frac{q}{4\pi\varepsilon_0 a\left(\frac{5}{4}-\cos\theta\right)^{1/2}} + \Phi_0$$

$$\Phi_0 = V_0 + V_0\cos\theta - \frac{q}{4\pi\varepsilon_0 a\left(\frac{5}{4}-\cos\theta\right)^{1/2}}$$

$$\Phi = V_0 + V_0\cos\theta - \frac{q}{4\pi\varepsilon_0 a\left(\frac{5}{4}-\cos\theta\right)^{1/2}} + \frac{q}{4\pi\varepsilon_0\left(r^2+a^2-ra\cos\theta\right)^{1/2}}$$

b.)

The greens functions is defined in jackson that a is radius of inner sphere and b is radius of outer sphere. To make since of problem I'm going to say b is inner radius and a is outer radius to actually will have $b \to 0$

$$G(\vec{x},\vec{x}') = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{Y_{lm}^* Y_{lm}}{(2l+1)\left[1-\left(\frac{b}{a}\right)^{2l+1}\right]} \left(r_<^l - \frac{b^{2l+1}}{r_<^{l+1}}\right)\left(\frac{1}{r_>^{l+1}} + \frac{r_>^l}{a^{2l+1}}\right)$$

$$G = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{Y_{lm}^* Y_{lm}}{(2l+1)} \left(r_<^l\right)\left(\frac{1}{r_>^{l+1}} + \frac{r_>^l}{a^{2l+1}}\right)$$

- azimuthal symmetry $m = 0$

$$G = 4\pi \sum_{l=0}^{\infty} \frac{Y_{l0}^* Y_{l0}}{(2l+1)} \left(r_<^l\right)\left(\frac{1}{r_>^{l+1}} + \frac{r_>^l}{a^{2l+1}}\right)$$

$r < a$:

- $r_<$ is r
- i.e. $r_>$ is a

$$G = 4\pi \sum_{l=0}^{\infty} \frac{Y_{l0}^* Y_{l0}}{(2l+1)} \left(r^l\right)\left(\frac{1}{a^{l+1}} + \frac{r^l}{a^{2l+1}}\right)$$

Use Jackson 3.128:

$$\Phi(\vec{x}) = \sum_{l}^{\infty} \left[\int \Phi(a,\theta') Y_{l0}^{*} \, d\Omega' \right] \left(\frac{r}{a}\right)^{l} Y_{l0}$$

$P_0 = 1$ $\quad Y_{l0} = \sqrt{\frac{2l+1}{4\pi}} P_l(\cos\theta)$

$P_1 = \cos\theta$

Solution is defined at $l = 0, 1$

$$\Phi = \int_0^{2\pi}\int_0^{\pi} V_0(1+\cos\theta') \sqrt{\frac{2(0)+1}{4\pi}} P_0(\cos\theta') \sin\theta' \, d\theta' d\phi' \sqrt{\frac{2(0)+1}{4\pi}} P_0(\cos\theta)$$

$$+ \int_0^{2\pi}\int_0^{\pi} V_0(1+\cos\theta') \sqrt{\frac{3}{4\pi}} P_1(\cos\theta') \sqrt{3/4\pi} \, P_1(\cos\theta) \left(\frac{r}{a}\right) \sin\theta' \, d\theta' d\phi'$$

$$\Phi = \frac{V_0 \cos\theta r + V_0}{a}$$

$l = 0, 1$

$\Phi =$

$$\sum_{l=0}^{\infty} \int Y_{lm}^{*}(\theta,\phi)\, Y_{lm}(\theta',\phi')\, d\Omega' \int r^{l} Y_{lm}$$

I_1

$$\Phi = \int_0^{2\pi}\int_0^{\pi} \frac{V_0}{4\pi} \sin\theta' d\theta' d\phi' + \int_0^{2\pi}\int_0^{\pi} \frac{V_0}{4\pi} \cos\theta' \sin\theta' d\theta' d\phi' \Big] I_2$$

$$+ \int_0^{2\pi}\int_0^{\pi} \frac{3V_0}{4\pi} \cos\theta \sin\theta' d\theta' d\phi' \frac{r}{a} \Big] I_3$$

$$+ \int_0^{2\pi}\int_0^{\pi} \frac{3V_0}{4\pi} \cos\theta \cos\theta'^{2} \sin\theta' d\theta' d\phi' \frac{r}{a} \Big] I_4$$

$$I_1 = \frac{V_0}{4\pi}(2\pi)\int_0^{\pi} \sin\theta' d\theta' = \frac{V_0}{4\pi} 4\pi = V_0$$

$$I_2 = \frac{V_0}{2}\int_0^{\pi} \cos\theta' \sin\theta' d\theta' \qquad u = \sin\theta' \quad du = \cos\theta' d\theta'$$

$$I_2 = 0, \quad I_3 = 0$$

$$I_4 = \frac{3V_0}{2}\cos\theta \frac{r}{a}\int_0^{\pi} \cos\theta'^{2} \sin\theta' d\theta' = \frac{3V_0}{2}\cos\theta \frac{2}{3}\frac{r}{a}$$

$$I_4 = V_0 \cos\theta \frac{r}{a}$$

$$\Phi = \frac{V_0 \cos\theta r + V_0}{a}$$

$l = 0, 1$

$\Phi =$

4.)

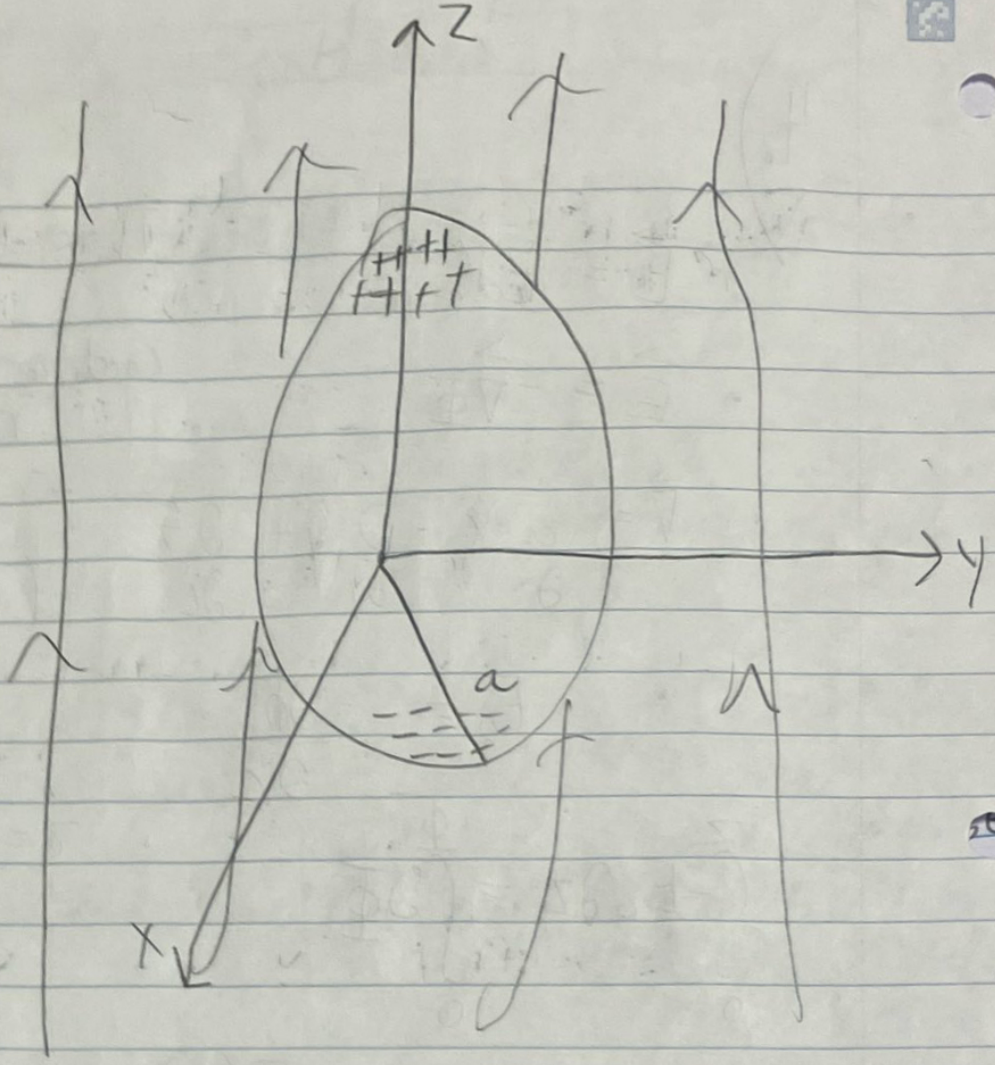
- $\vec{E} = E_0 \hat{z}$
- $\vec{E} = -\vec{\nabla}\Phi$
- $\vec{\nabla} = \frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}$ $\left((r, \theta, \phi)\right)$
- $E_0 = -\frac{\partial \Phi}{\partial z}$
- $\int_0^z -E_0 \partial z = \int_0^{\Phi} \partial \Phi$ · $z = r\cos\theta$
- $\Phi(z) = -E_0 z$
- $\Phi(r \to \infty, \theta) = -E_0 r\cos\theta$

This is a boundary condition when $r \to \infty$.

Boundary conditions:

1) $\Phi(r \gg a, \theta) = -E_0 r\cos\theta$
2) $\Phi(r = a, \theta) = 0$

$$\Phi = \int_0^{2\pi}\int_0^{\pi} \frac{V_0}{} \sin\theta' d\theta' d\phi'$$



- $\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial \Phi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial \Phi}{\partial\theta}\right)$

$$+ \frac{1}{r^2\sin^2\theta}\frac{\partial^2 \Phi}{\partial\phi^2}$$

- $\nabla^2\Phi = 0$

- $\Phi = \frac{V(r)}{r} P(\theta)Q(\phi)$

$$PQ\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial U}{\partial r}\right)+\frac{UQ}{r^2\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dP}{d\theta}\right)$$

$$+\frac{UP}{r^2\sin^2\theta}\frac{d^2Q}{d\phi^2}=0$$

$$\frac{1}{r^2}\frac{1}{U}\frac{\partial}{\partial r}\left(r^2\frac{\partial U}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{1}{P}\frac{d}{d\theta}\left(\sin\theta\frac{dP}{d\theta}\right)$$

$$+\frac{1}{r^2\sin^2\theta}\frac{1}{Q}\frac{\partial^2Q}{\partial\phi^2}=0$$

* multiply by r^2 *

$$\frac{1}{U}\frac{\partial}{\partial r}\left(r^2\frac{\partial U}{\partial r}\right)+\frac{1}{\sin\theta}\frac{1}{P}\frac{d}{d\theta}\left(\sin\theta\frac{dP}{d\theta}\right)$$

$$+\frac{1}{\sin^2\theta}\frac{1}{Q}\frac{\partial^2Q}{\partial\phi^2}=0$$

$$\frac{1}{Q}\frac{\partial^2Q}{\partial\phi^2}=-m^2$$

$$Q(\phi)=Ae^{im\phi}+Be^{-im\phi}$$

(for this problem $m=0$)

$\Phi =$

$$\underbrace{\frac{1}{U}\frac{\partial}{\partial r}\left(r^2\frac{\partial U}{\partial r}\right)}_{l(l+1)} + \frac{1}{\sin\theta}\frac{1}{P}\frac{d}{d\theta}\left(\sin\theta\frac{dP}{d\theta}\right) = 0$$

$$\frac{1}{U}\frac{\partial}{\partial r}\left(r^2\frac{\partial U}{\partial r}\right) = l(l+1)$$

$$\frac{1}{U}\left(2r\frac{\partial U}{\partial r} + r^2\frac{\partial^2 U}{\partial r^2}\right) = l(l+1)$$

$$r^2\frac{\partial^2 U}{\partial r^2} + 2r\frac{\partial U}{\partial r} - l(l+1)U = 0$$

$U = r^k$

$U' = kr^{k-1}$

$U'' = k(k-1)r^{k-2}$

$$k(k-1) + 2k - l(l+1) = 0$$

$$k^2 - k + 2k - l(l+1) = 0$$

$$k^2 + k - l(l+1) = 0$$

$$k(k+1) - l(l+1) = 0$$

$$k(k+1) = l(l+1)$$

$k = l$

$k = -l-1$

- $U(r) = A_\ell r^{\ell} + B_\ell r^{-(\ell+1)}$

- Legendre equation:

) $P\,\ell(\ell+1) + \frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dP}{d\theta}\right) = 0$

- $P(\theta) = P_\ell(\cos\theta)$

- $\Phi(r,\theta) = \sum_{\ell=0}^{\infty}\left[A_\ell r^{\ell} + B_\ell r^{-(\ell+1)}\right]P_\ell(\cos\theta)$

Boundary conditions:

- $\Phi(r=a,\theta) = 0$ (1)
- $\Phi(r\to\infty,\theta) = -E_0 r\cos\theta$ (2)

apply boundary condition # 1:

$A_\ell a^{\ell} + \frac{B_\ell}{a^{\ell+1}} = 0$, $B_\ell = -A_\ell a^{2\ell+1}$

$$\Phi(r,\theta) = \sum_{\ell=0}^{\infty} \left(A_\ell r^\ell - \frac{A_\ell a^{2\ell+1}}{r^{\ell+1}} \right) P_\ell(\cos\theta)$$

) apply boundary condition #2.

- Get rid of 2nd term of parentesis.

$$-E_0 r\cos\theta = \sum_{\ell=0}^{\infty} A_\ell r^\ell P_\ell(\cos\theta)$$

<u>orthonormality:</u>

$$\int_{-1}^{1} P_{\ell'}(x) P_\ell(x)\, dx = \frac{2}{2\ell+1}\delta_{\ell'\ell}$$

$$\int_0^{\pi} P_{\ell'}(\cos\theta) P_\ell(\cos\theta) \sin\theta\, d\theta \qquad x = \cos\theta, \quad dx = -\sin\theta\, d\theta$$

$$= \frac{2}{2\ell+1}\delta_{\ell'\ell}$$

- $P_0(\cos\theta) = 1$
- $P_1(\cos\theta) = \cos\theta$
- $P_2(\cos\theta) = \left(\frac{3\cos^2\theta - 1}{2}\right)$

orthonormality for $l = 1$:

- multiply both sides by $\int_0^\pi P_{l'}(\cos\theta)\sin\theta\, d\theta$

- $-\int_0^\pi E_0 P_l(\cos\theta)\sin\theta\, r\cos\theta\, d\theta = A_{el}\left(\frac{2}{2l+1}\right)\delta_{ll'}$

$$A_l = -E_0\left(\frac{2l+1}{2r^{1-l}}\right)\int_0^\pi P_l(\cos\theta)\sin\theta\cos\theta\, d\theta$$

$l = 1$:

$P_1(\cos\theta) = \cos\theta$

$$A_1 = -\frac{3E_0}{2}\int_0^\pi \cos^2\theta\sin\theta\, d\theta$$

$u = \sin\theta$

$du = \cos\theta\, d\theta$

$$A_1 = -\frac{3E_0}{2}\left(\frac{2}{3}\right) = -E_0$$

- every other coefficient doesn't match or is zero.

$$\Phi_{out}(r,\theta) = -E_0 \cos\theta \left(r - \frac{a^3}{r^2} \right)$$

- $\vec{E} = -\vec{\nabla}\Phi$

- $\vec{\nabla}_{sphere} = \frac{\partial}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial}{\partial \theta}\hat{\theta} + \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi}\hat{\phi}$

- $\vec{\nabla}\Phi = -E_0 \cos\theta \frac{\partial}{\partial r}\left(r - \frac{a^3}{r^2} \right)\hat{r}$

$$\frac{-1}{r} E_0 \left(r - \frac{a^3}{r^2} \right) \frac{\partial}{\partial \theta}(\cos\theta)\,\hat{\theta}$$

- $= -E_0 \cos\theta \left(1 + \frac{2a^3}{r^3} \right)\hat{r} + \frac{1}{r} E_0 (r - a^3/r^2)\sin\theta\,\hat{\theta}$

$$\vec{E} = E_0 \cos\theta \left(1 + 2a^3/r^3\right)\hat{r} - \frac{1}{r} E_0 \left(r - \frac{a^3}{r^2} \right)\sin\theta\,\hat{\theta}$$